

20250210EB_MN

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2025年2月10日 星期一

do not use due to the bleedthrough from the microscope

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年2月12日 星期三

T2 -> T3

2025年2月14日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年2月17日 星期一

T4 -> T5

2025年2月19日 星期三

T5 -> T6

2025年2月21日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年2月24日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatant (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

Table3					
	A	B	C	D	E
1	conc. (M/mL)	total (4mL)	plate A & plate B C1,2, D1,2	the rest of plate B	plate C
2	1.51	6.04	~250K	~200K	428K

*note: I didn't notice the concentration was not trypan blue corrected. Therefore, I seeded 428K/well. Then I re-plated 6 wells of plate A into plate A +4 wells of plate B (6 wells into 10 wells) and re-plated 6 wells of plate B into 12 wells in plate B.

plateA (vLH92)						
	1	2	3	4	5	6
A	BP	+astro BP	+astro +TIC BP			
B	hypo	+astro hypo	+astro +TIC hypo			
C						
D						

plateB

	1	2	3	4	5	6
A	vLH110 (didn't image at the end)					
B	vLH111 (didn't image at the end)					
C	vLH92 (4) BP (5) BP+4AP 100uM	vLH92 (7) BP+TTX 50nM	BP+DMSO	BP+Thiamet-% 10uM		
D	vLH92 (6) hypo+4AP 100uM	vLH92 (8) hypo+TTX 50nM	hypo+DMSO	hypo+Thiamet-G 10uM		

plateC

	1	2	3	4	5	6
A						
B					vLH129	
C					vLH129	
D						

2025年2月25日 星期二

Change T7 -> T7

2025年2月27日 星期四

T7 -> BP T8

2025年2月28日 星期五

Gave plate A to Fra to add astrocyte

2025年3月1日 星期六

-200uL +300uL T8

- cells on plate B with only 200K at the beginning didn't look well

2025年3月3日 星期一

200uL BP T8 -> 300 uL BP T8

treat 0.1 ug/mL dox to plate C

2025年3月4日 星期二

plate C didn't express ER-GEM -> add 1ug/mL dox instead

A1-4&B1-4 on plateB still look bad, May consider not using them. C3,4 and D3,4 didn't look well, but they are better than row A & B

2025年3月5日 星期三

200uL BP T8 -> 300 uL BP T8
plate C express ER-GEM!

2025年3月6日 星期四

Imaging

O.C. 16bit, 600x900, 60x				^
	A	B	C	
1	GFP	30%	200ms	
2	Dapi-GFPem	40%	500ms	
3	Cy5	50%	200ms	
4	DsRed	50%	500ms	

*** note1: for plate AA3 1h, B3 1h, plateB C1 BP, I use dual camera to image GFP&Cy5 and Dapi-GFPem&DsRED together --> HOWEVER, there is bleedthrough from GFP to Cy5 and DsRED to Dapi-GFPem

*** note2: I noticed the bleedthrough when I imaged DsRED to Dapi-GFPem. Therefore, I separate them when I imaged B21h --> HOWEVER, there is bleedthrough from GFP to Cy5, which I didn't noticed at that time

***note3: the rest of images are all separated channels

***note4: For plateA, every well treated astrocyte supplement even without astrocytes

Table1



	A	B	C	D
1	plate A			plate B
2	10:00	A3&B3 TIC in BP		
3	15:03	B3 BP + TIC + JFX650 (1:2500) + SPY555		
4	15:30	A3 hypo + TIC + JFX650 (1:2500) + SPY555		
5	16:23	image A3 (6h TIC)		
6	16:45	image B3 (6h TIC)		C1 BP+JFX+SPY
7	18:12	A2 BP +JFX +SPY+H		
8	18:26			image C1 BP only
9	18:30	B2 hypo +JFX +SPY+H		
10	18:40	A1 BP +JFX +SPY+H		
11	18:50	B1 hypo +JFX +SPY+H		
12	19:30	image A2		
13	19:45	image B2		
14	19:58	image A1		
15	20:00			C1 BP + 4AP 100uM
16	20:12	image B1		
17	20:17			D1 hypo + 4AP 100uM
18	20:30			C2 BP + TTX 50nM
19	20:42			D2 hypo + TTX 50nM
20	20:57			image C1
21	21:13			image D1
22	21:30			image C2
23	21:42			image D2

2025年3月7日 星期五

Image Plate A 24h and Plate B C3,4&D3,4

*It was really hard to find good cellse in Plate B C3,4&D3,4 -> decide not to image row A&B in plateB

*note1: only 14 positions for MN_noA_BP_24h

*note2: Plate A 24h: Dapi-GFPem channel somehow has the pattern of Halo or SPY595, especially with BP

O.C. for plate B C3,4,D3,4: 16bit, 704x704, 60x				^
	A	B	C	
1	BF		100ms	
2	Cy5	30%	200ms	
3	DsRed	30%	500ms	

Table2				
	A	B	C	D
1	plate A			plate B
2	11:30	treat A3 & B3 with media+JFX650		
3	12:41	image B3 (24h)		
4	12:56	image A3 (24h)		
5	13:17			C3 BP+DMSO
6	13:50			D3 hypo+DMSO
7	14:05			C4 BP+Thiamet
8	14:20			D4 hypo+Thiamet
9	14:44			image C3
10	15:00			image D3
11	15:15			image C4
12	15:30			image D4
13	15:40	treat A1,2 & B1,2 with media+JFX650		
14	17:12	image A2 (24h)		
15	17:30	image B2 (24h)		
16	17:55	image A1 (24h)		
17	18:24	image B1 (24h)		

summary of files that may not be used due to bleedthrough

- plateA:
- A3 1h, B31h, B21h (TDP43&ADP)
- A1-3, B1-3 24h (ADP)
- plateB:
- C1 BP1h (TDP43&ADP)